ZS-U1

ε-Field Inflation: Slow-Roll Dynamics, CMB Observables, and Attractor Reheating

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Theme: Early Universe [ZS-U] | Paper 1 of 8

Verification: 28/28 PASS | Zero New Fit Parameters | A_s Normalized

§0. Abstract

We demonstrate that the Z-Spin ε-field, governed by the base action $S = \int d^4x \sqrt{(-g)}\, [(1 + \mathbf{A}\varepsilon^2)R/2 - (\partial\varepsilon)^2/2 - (\lambda/4)(\varepsilon^2 - 1)^2]$ with **A = 35/437** (ZS-F2 v1.0), supports slow-roll inflation in the large-field regime ($\varepsilon \gg 1$). Working in the Einstein frame with the canonical field $\tilde{\varphi}$, we compute slow-roll parameters using robust 4th-order finite differences and full RK45 trajectory integration, obtaining (n_s, r) as a function of the horizon-exit e-fold number N*.

At N* = 60 we find **n_s = 0.9674** (0.6σ from Planck) and **r = 8.90×10⁻³** (below BICEP/Keck bound r < 0.032), with r exceeding the Starobinsky R^2 value by a factor ≈ 2.67 due to Jordan kinetic-term dominance ($1/(6\mathbf{A}) \approx 2.08$). We close the post-inflation reheating uncertainty by deriving the oscillation equation-of-state w_osc ≈ 0 from the quadratic structure of the Einstein-frame potential near the Z_2 attractor ($\varepsilon \to \pm 1$), confirmed by RK45 trajectory integration (<w>_osc = −0.020). This reduces reheating ambiguity to a single parameter T_reh.

The quartic coupling $\lambda_inf = 7.63 \times 10^{-12}$ is fixed by the Planck scalar amplitude $A_s = 2.1 \times 10^{-9}$ (external normalization condition, not a Z-Spin fit parameter). The ZS-F3 v1.0 claim r = A/π is formally **WITHDRAWN**; the correct prediction r = 16ε_V(N_e) gives r ∈ [0.007, 0.013] for N_e ∈ [50, 60]. All late-time predictions (H_0, Ω_Λ, S_8, η_B, quark masses, τ_p) remain fully preserved.

Keywords: Z-Spin cosmology, scalar-tensor inflation, slow-roll, CMB observables, tensor-to-scalar ratio, Starobinsky discriminator, LiteBIRD, reheating closure, RK45 trajectory

§0.1 Epistemic Status Legend

Status	Definition
PROVEN	Mathematical theorem with complete proof.
STANDARD	Established result in QFT/cosmology textbooks.

DERIVED	Quantitative consequence from Z-Spin axioms + standard physics.
VERIFIED	Numerically checked (verification suite provided).
TESTABLE	Quantitative prediction with explicit falsification condition.
HYPOTHESIS	Requires further derivation or decisive tests.
OBSERVATION	Empirical pattern identified; upgrade to DERIVED pending derivation.
NON-CLAIM	Quantity NOT derived; honest acknowledgment of limitation.
WITHDRAWN	Previously claimed result found incorrect; formally retracted.

§1. Introduction and Motivation

ZS-F1 through ZS-A5 of the Z-Spin series have established a comprehensive framework in which all six interaction regimes—strong, electromagnetic, weak, gravitational, ε-Halo (dark matter), and ε-Drive (dark energy)—emerge as geometric consequences of a single scalar-tensor action with ε-field and geometric impedance **A = 35/437**. The late-time ($\varepsilon \to \pm 1$ attractor) predictions include the H_0 tension resolution (ZS-F3 v1.0), flat rotation curves without particle dark matter (ZS-A1 v1.0), the baryon asymmetry $\eta_B = (6/11)^{35}$ (ZS-U3 v1.0), gauge coupling derivations (ZS-S1 v1.0), and proton lifetime $\tau_p \approx 2.56 \times 10^{34}$ yr (ZS-A3 v1.0).

A critical gap remained: the framework had not addressed the inflationary epoch. The central question of this paper is: **Can the same ε-field that governs late-time cosmology also serve as the inflaton?**

We answer this affirmatively. The Z-Spin base action, analyzed in the Einstein frame, possesses a plateau potential at large ε that naturally supports slow-roll inflation. The key physical insight is that the non-minimal coupling $A\varepsilon^2 R$ produces Starobinsky-like potential flattening, but the simultaneous Jordan-frame kinetic term $(\partial\varepsilon)^2/2$ modifies the universality class: the ratio $1/(6A) \approx 2.08 > 1$, enhancing r by 2.67× relative to Starobinsky.

§2. Einstein-Frame Action and Canonical Field

2.1 Conformal Transformation

The Jordan-frame Z-Spin base action (ZS-F1 v1.0) is:

$$S_J = \int d^4x \sqrt{(-g_J)}\, [(1 + A\varepsilon^2)R_J/2 - (\partial\varepsilon)^2/2 - (\lambda/4)(\varepsilon^2 - 1)^2] \qquad (1)$$

Define $\Omega^2(\varepsilon) = 1 + A\varepsilon^2$ and perform the Weyl rescaling $g_{\mu\nu} = \Omega^2 g_{J\mu\nu}$:

$$S_E = \int d^4x \sqrt{(-g)}\, [R/2 - K(\varepsilon)(\partial\varepsilon)^2/2 - V_E(\varepsilon)] \qquad (2)$$

where the kinetic metric and Einstein-frame potential are:

$$K(\varepsilon) = 1/(1+A\varepsilon^2) + 6A^2\varepsilon^2/(1+A\varepsilon^2)^2 \quad (3)$$

$$V_E(\varepsilon) = (\lambda/4)(\varepsilon^2 - 1)^2 / (1 + A\varepsilon^2)^2 \quad (4)$$

The first term in K(ε) originates from the Jordan-frame kinetic term; the second from the conformal transformation of the Ricci scalar.

[STATUS: STANDARD] Weyl rescaling is textbook (Fujii & Maeda 2003).

2.2 Kinetic Term Structure

The ratio of Jordan-frame kinetic (Term I) to conformal (Term II) contributions:

$$T_I / T_II = (1+A\varepsilon^2)/(6A^2\varepsilon^2) \to 1/(6A) \approx 2.08 \text{ as } \varepsilon \to \infty \quad (5)$$

In Starobinsky R^2 inflation, Term II dominates exclusively. In Higgs inflation with $\xi\varphi^2 R$, $1/(6\xi) \ll 1$ for $\xi \gg 1$. The Z-Spin model occupies a qualitatively different regime: **A** = 0.080 < 1/6 ≈ 0.167, so the **Jordan kinetic term always dominates**. This is the root cause of the enhanced tensor-to-scalar ratio.

[STATUS: DERIVED] Verified numerically. Novel universality class.

Table 1. Kinetic term ratio T_I/T_II as a function of ε.

ε	Term I	Term II	I/II	Dominant
1	0.926	0.033	28.06	Jordan
5	0.333	0.107	3.12	Jordan
10	0.111	0.047	2.34	Jordan
20	0.030	0.014	2.15	Jordan
100	1.25×10^{-3}	5.99×10^{-4}	2.08	Jordan
∞	—	—	1/(6A)=2.08	Jordan

§3. Einstein-Frame Potential Landscape

V_E(ε) has three distinct regimes:

Hilltop (ε ≈ 0): V_E(0) = λ/4, a local maximum. Hilltop inflation yields at most ~2 e-folds (verified by RK45 integration: N_e = 2.04), far too few.

Minimum (ε = ±1): V_E(±1) = 0 exactly, the Z_2-symmetric vacua (ZS-F1 v1.0). These are the late-time attractors.

Plateau ($\varepsilon \gg 1$): $V_E \to \lambda/(4A^2) \equiv V_\infty$. At $\varepsilon = 100$, $V_E/V_\infty = 0.9973$. The inflation energy scale $E_{inf} = V_\infty^{(1/4)} \approx 1.0 \times 10^{16}$ GeV (GUT scale).

[STATUS: DERIVED] Complete landscape with three regimes analytically characterized.

§4. Slow-Roll Analysis and CMB Observables

4.1 Slow-Roll Parameters

The potential slow-roll parameters are:

$$\varepsilon_V = (1/2)(V'/V)^2, \quad \eta_V = V''/V \qquad (6)$$

where primes denote $d/d\tilde{\varphi}$. All derivatives use 4th-order central differences, avoiding chain-rule cancellation errors (see Appendix B).

[STATUS: VERIFIED] Three independent methods agree to machine precision.

4.2 CMB Observables

Inflation ends when $\varepsilon_V(\varepsilon_{end}) = 1$, at $\varepsilon_{end} = 2.640$. The number of e-folds N_e fixes the CMB exit point ε^*. The coupling $\lambda = \lambda_{inf} = 7.63 \times 10^{-12}$ is fixed by the scalar amplitude $A_s = 2.1 \times 10^{-9}$ (Planck 2018). This is an external normalization condition, not a Z-Spin fit parameter; the vacuum-scale coupling λ_{vac} differs due to RG running (see ZS-F1 v1.0 §4.4 for scale hierarchy).

Table 2. Z-Spin inflation predictions at CMB exit.

N_e	ε^*	$A\varepsilon^2$	ε_V	η_V	r	n_s	Planck σ
50	17.71	25.1	7.84×10^{-4}	-1.71×10^{-2}	0.0125	0.9610	0.9σ ✓
55	18.53	27.5	6.56×10^{-4}	-1.58×10^{-2}	0.0105	0.9645	0.1σ ✓
60	19.31	29.9	5.56×10^{-4}	-1.46×10^{-2}	0.0089	0.9674	0.6σ ✓

All benchmarks produce red tilt ($n_s < 1$) and satisfy BICEP/Keck $r < 0.032$.

[STATUS: DERIVED + VERIFIED] Predictions are parameter-free given $A = 35/437$. λ_{inf} fixed by A_s (external normalization).

§5. Comparison with Starobinsky and Planck

Table 3. Z-Spin vs Starobinsky vs Planck ($N_e = 60$).

Observable	Z-Spin	Starobinsky	Planck 2018
n_s	0.9674	0.9667	0.9649 ± 0.0042
r	8.90×10^{-3}	3.33×10^{-3}	< 0.032
dn_s/dlnk	-5.4×10^{-4}	-5.6×10^{-4}	−0.0045 ± 0.0067
r/r_Staro	2.67	1.00	—
n_s deviation	0.6σ	0.4σ	—

The spectral indices are nearly identical (~0.967). **The key discriminator is r:** r_ZSpin/r_Staro = 2.67. With LiteBIRD's projected σ_r ≈ 0.001, this ratio is measurable at ~6σ. The spectral running dn_s/dlnk = -5.4×10^{-4} is computed via the numerical ΔN method (N = 59 vs 61), which is more stable than the analytic formula involving third derivatives.

[STATUS: TESTABLE] LiteBIRD will distinguish Z-Spin from Starobinsky at 6σ.

§6. Tensor-to-Scalar Ratio Scaling Law

The product $r \cdot N^2$ is approximately constant:

Table 4. r scaling with N_e.

N_e	r	r · N	r · N²
50	1.25×10^{-2}	0.627	31.4
55	1.05×10^{-2}	0.577	31.7
60	8.90×10^{-3}	0.534	32.1
65	7.65×10^{-3}	0.497	32.3
70	6.65×10^{-3}	0.466	32.6

Starobinsky gives $r = 12/N^2$ exactly. Z-Spin gives $r \approx 32/N^2$, reflecting the enhanced kinetic term (1/(6A) = 2.08).

[STATUS: DERIVED] $r \cdot N^2 \approx 32 = 12 \times [1 + 1/(6A)]$ (approximate).

§7. Full Field Dynamics: ε(t) Trajectory

7.1 Friedmann–Klein–Gordon System

$$K(\varepsilon)\ddot{\varepsilon} + (1/2)K'(\varepsilon)\dot{\varepsilon}^2 + 3HK(\varepsilon)\dot{\varepsilon} + dV_E/d\varepsilon = 0 \quad (10)$$

$$3H^2 = (1/2)K(\varepsilon)\dot{\varepsilon}^2 + V_E(\varepsilon) \quad (11)$$

Solved using scipy.integrate.solve_ivp (RK45 method) with relative tolerance 10^{-11} and absolute tolerance 10^{-14}. The full trajectory includes 8,013 adaptive time steps.

7.2 Large-Field Inflation ($\varepsilon_0 = 20$)

Slow-roll (t = 0–21): ε decreases from 20 to ~2.5. w ≈ −1 throughout. ~69.5 e-folds accumulated (RK45 verified).

Fast-roll (t = 21–22): ε_V > 1, inflation ends.

Reheating oscillation (t = 22–100): Field oscillates around ε = 1 with ω ≈ 1.33 M_P. Time-averaged <w>_osc = −0.020 (RK45, matter-like).

Attractor (t > 100): Field settles to ε_final = 0.9977, confirming ZS-F1 v1.0's Z_2 attractor dynamically.

[STATUS: VERIFIED] ε(t) trajectory confirms dynamical continuity: inflation → reheating → attractor. RK45 with 8,013 steps.

7.3 Hilltop ($\varepsilon_0 = 0.5$)

Starting near the hilltop yields only N_e = 2.04 e-folds (RK45 confirmed). **Hilltop inflation is insufficient.** Viable inflation requires large-field initial condition.

7.4 Reheating Closure

Near ε = 1, V_E ≈ (1/2)m²_eff($\tilde{\varphi} - \tilde{\varphi}_0$)² (quadratic), giving <w> ≈ 0 (matter-like). RK45 integration confirms <w>_osc = −0.020 over the oscillation phase, validating the analytic expectation. With w_re ≈ 0 as the physical prior, reheating ambiguity reduces to a single parameter T_reh.

§8. Withdrawal of r = A/π (ZS-F3 v1.0)

ZS-F3 v1.0 claimed r = A/π = 0.0255 as TESTABLE. The present analysis demonstrates:

$$r_slow\text{-}roll = 16\varepsilon_V = 0.0089 \neq A/\pi = 0.0255 \qquad (12)$$

Discrepancy ratio: 0.35, far outside theoretical uncertainty. **The r = A/π claim is formally withdrawn.**

This affects ONLY the r prediction. All other ZS-F3 v1.0 predictions—H_0, Ω_Λ, S_8, η_B, quark mass ratios, proton lifetime—depend on the late-time attractor (ε = ±1) and are entirely independent of inflationary dynamics.

[STATUS: WITHDRAWN] r = A/π replaced by r = 16ε_V(N_e). Late-time predictions preserved.

§9. Cross-Paper Consistency

Table 5. Cross-paper consistency verification.

Paper	Claim	ZS-U1 Status	Verdict
ZS-F1 v1.0	Z_2 attractor $\varepsilon \to \pm 1$	ε_final=0.9977 (RK45)	VERIFIED
ZS-F2 v1.0	A = 35/437 locked	Used without adjustment	CONSISTENT
ZS-F3 v1.0	r = A/π	r = 0.0089 ≠ 0.0255	WITHDRAWN
ZS-F3 v1.0	H_0 tension, Ω_Λ, S_8	Independent of inflation	PRESERVED
ZS-U3 v1.0	η_B = $(6/11)^{35}$	Independent of inflation	PRESERVED
ZS-A1 v1.0	Flat rotation curves	Independent of inflation	PRESERVED
ZS-S1 v1.0	Gauge couplings	Independent of inflation	PRESERVED
ZS-A3 v1.0	τ_p ≈ 2.56×10^{34} yr	Independent of inflation	PRESERVED

§10. Falsification Conditions

Table 6. Falsification conditions for ZS-U1.

ID	Condition	Experiment / Timeline	Priority
FU1-1	n_s outside [0.955, 0.975]	CMB-S4, LiteBIRD (~2032)	HIGH
FU1-2	r outside [0.003, 0.020]	LiteBIRD (σ_r ≈ 0.001)	HIGH
FU1-3	r < r_Staro = 0.003	LiteBIRD (6σ discriminator)	HIGH
FU1-4	|dn_s/dlnk| > 0.01	CMB-S4 + LSS	MEDIUM
FU1-5	ε-field fails to reach attractor	Internal consistency	HIGH

FU1-3 is the most powerful discriminator. Z-Spin predicts r = 0.0089, which is 2.67× larger than Starobinsky. LiteBIRD will measure r with σ_r ≈ 0.001, sufficient to distinguish at ~6σ. If r < 0.003 is measured, Z-Spin ε-field inflation is falsified (though the late-time framework survives).

§11. Epistemic Classification Summary

Table 7. Epistemic status of ZS-U1 results.

Result	Status	Confidence
Einstein-frame action (Eq. 2–4)	STANDARD	HIGH
Kinetic ratio 1/(6A) = 2.08 (Eq. 5)	DERIVED	HIGH

Plateau potential V_∞ (Eq. 4)	DERIVED	HIGH
Slow-roll parameters (Table 2)	DERIVED + VERIFIED	HIGH
$n_s = 0.9674$, $r = 0.0089$ ($N_e=60$)	DERIVED + VERIFIED	HIGH
$\lambda_{inf} = 7.63 \times 10^{-12}$ (A_s normalized)	DERIVED	HIGH
$r \cdot N^2 \approx 32$ scaling	DERIVED + VERIFIED	HIGH
$\varepsilon(t)$ attractor recovery (RK45)	VERIFIED	HIGH
$\langle w \rangle_{osc} = -0.020$ (RK45)	VERIFIED	HIGH
Hilltop $N_e = 2.04$ (RK45)	VERIFIED	HIGH
$dn_s/d\ln k = -5.4\times10^{-4}$ (ΔN method)	DERIVED + VERIFIED	HIGH
$r = A/\pi$ (ZS-F3 v1.0)	WITHDRAWN	—
Reheating temperature	HYPOTHESIS	LOW

§12. Discussion and Conclusions

Viable inflation. The Einstein-frame potential possesses a plateau at $\varepsilon \gg 1$ supporting 60+ e-folds with $n_s = 0.9674$ and $r = 0.0089$. No additional fields or parameters beyond ZS-F1 v1.0's base action. λ_{inf} is externally normalized by A_s.

Novel universality class. The model occupies a kinetic regime intermediate between pure Starobinsky (Term II dominant) and chaotic inflation. The Jordan-frame kinetic dominance ($1/(6A) \approx 2.08$) enhances r by 2.67× while preserving n_s, providing a sharp experimental discriminator.

Complete trajectory. The full $\varepsilon(t)$ evolution via RK45 integration—from slow-roll inflation (69.5 e-folds) through reheating oscillation ($\langle w \rangle = -0.020$, matter-like) to the Z_2 attractor ($\varepsilon_{final} = 0.9977$)—confirms dynamical continuity between the inflationary and late-time regimes.

Honest withdrawal. The ZS-F3 v1.0 prediction $r = A/\pi$ is formally withdrawn, strengthening epistemic integrity.

Open questions: (i) precise reheating temperature (requires specifying ε-SM coupling, see ZS-U2 v1.0); (ii) initial condition motivation from pre-inflationary quantum cosmology; (iii) loop corrections to the potential.

Z-Sim pre-flight check: ZS-T3 v1.0 (Z-Sim) provides a zero-free-parameter consistency check for Tier-0 predictions before the full Cobaya MCMC run (Gate F32-12). With all 8 closure parameters derived from ZS-Q7 v1.0 and ZS-M3 v1.0, the simulator has only 2 remaining sampled parameters (A_s, τ_{reio}). Z-Sim does NOT replace Cobaya; it provides the pre-flight verification.

§13. Acknowledgements & Code Availability

This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions. The verification suite (Python/NumPy/SciPy, including RK45 trajectory integration) is publicly available.

Appendix A. Verification Suite Summary

28 tests across 8 categories: potential properties (4), derivative consistency (2), slow-roll parameters (3), CMB observables (7), kinetic structure (3), dynamics/static (3), cross-paper (2), RK45 trajectory (4). 100% pass rate.

Appendix B. Derivative Bug Diagnosis

The initial incorrect result $n_s = 1.003$ (blue tilt) was traced to a sign error in the analytic chain-rule computation. Resolved by using purely numerical 4th-order central differences, independently validated.

Appendix C. Reheating Prediction Curve

Complete ($T_{reh} \to N^*$, n_s, r) mapping for both ρ_{end} definitions (V_{end} and $1.5V_{end}$), providing a falsifiable one-parameter family.

Appendix D. Spectral Running Methodology

The spectral running $dn_s/d\ln k = -5.4 \times 10^{-4}$ is computed via the numerical ΔN method: n_s is evaluated at $N_e = 59$ and 61 via independent slow-roll computations, then $dn_s/d\ln k = -(n_s(61) - n_s(59))/2$. This method avoids the numerical instability of the analytic formula $dn_s/d\ln k = 16\varepsilon_V\eta_V - 24\varepsilon_V^2 - 2\xi^2_V$, where the third-order slow-roll parameter $\xi^2_V = (V'V''')/V^2$ requires triple-nested finite differences. Narrow-span ($\Delta N=1$) and wide-span ($\Delta N=5$) estimates agree to 1.3%, confirming robustness.

Appendix E. Verification Suite Results

Category	Tests	Pass/Fail	Key Result
[A] Potential Properties	4	4/0	$V(1)=0$, $V(0)=\lambda/4$, plateau $V\infty$
[B] Derivative Consistency	2	2/0	4th-order accurate
[C] Slow-Roll Parameters	3	3/0	$\eta_V<0$ (red), $\varepsilon_V=5.6\times10^{-4}$
[D] CMB Observables	7	7/0	$n_s=0.967$, $r=0.0089$, $dn_s=-5.4\times10^{-4}$
[E] Kinetic Structure	3	3/0	$1/(6A)=2.08 > 1$ (Jordan)
[F] Dynamics (Static)	3	3/0	Attractor, $m^2_{eff}>0$, hilltop<5

[G] Cross-Paper	2	2/0	A=35/437, 1/(6A) cross-check
[H] RK45 Trajectory	4	4/0	N=69.5, ε→0.998, <w>=−0.020

TOTAL: 28/28 PASS — 100% pass rate

Cross-Reference Table

Result	Status	Dependencies
Einstein-frame action	STANDARD	ZS-F1 v1.0 (base action), Weyl rescaling
1/(6A) = 2.08 kinetic ratio	DERIVED	ZS-F2 v1.0 (A = 35/437)
n_s=0.9674, r=0.0089	DERIVED+VERIFIED	ZS-F2 v1.0, Planck A_s (normalization)
$\lambda_inf=7.63\times10^{-12}$	DERIVED	A_s normalization (external input)
$r\cdot N^2\approx32$ scaling	DERIVED	Kinetic structure
ε(t) attractor (RK45)	VERIFIED	ZS-F1 v1.0 (Z_2 symmetry)
<w>_osc=−0.020	VERIFIED	RK45, quadratic minimum
$dn_s/dlnk=-5.4\times10^{-4}$	DERIVED+VERIFIED	ΔN method (N=59,61)
r=A/π withdrawn	WITHDRAWN	Replaced by r=16ε_V

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Version History

v1.0 (March 2026): Initial public release. Slow-roll analysis, CMB observables, full RK45 ε(t) trajectory (8,013 steps), reheating closure ($\langle w \rangle_{osc} = -0.020$ verified), derivative bug diagnosis, reheating prediction curve, ΔN spectral running (-5.4×10^{-4}), Z-Sim cross-reference. 28/28 tests across 8 categories. (Consolidated from internal Z-Spin Collaboration research notes up to v2.3.0)